THE EQUIVARIANT STOLZ POSITIVE SCALAR CURVATURE EXACT SEQUENCE IS EXACT

MASSIMILIANO PUGLISI, THOMAS SCHICK, VITO FELICE ZENOBI

Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/geometry-fl04-stolz-positive-scalar-curvature-st/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the retained preamble is the source's own layout adapter, if any, plus the author's macro block at lines 11-108 of sources/geometry-fl04-stolz-positive-scalar-curvature-st/pinned-original-source.tex (everything between the publisher class metadata and \begin{document}, comments excluded). Nothing was added, no macro was invented, and the publisher document class is deliberately not reproduced; the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Metadata provenance: source_id and source_version were taken from sources/geometry-fl04-stolz-positive-scalar-curvature-st/license-observation.json; the title from SOURCE.json; the author list from the source metadata record; the source URL from the first non-empty of source_url, source_urls[0] and the evidence's fixed_version_url/license_url. The licence evidence file used by this card is sources/geometry-fl04-stolz-positive-scalar-curvature-st/license-observation.json, recorded in license.evidence_path so that the binding target is explicit.

Editorial note. External references are navigation pointers only; every deliberately kept step of the author's argument stays inside the counted spans. The displayed numbers of those pointers are the author's own numbering under the author's \newtheorem declarations (section.index as declared), computed mechanically by the lane's scanner.

Editorial note. Bibliography provenance: the source stages its bibliography inside the pinned original (its bibliography block starts at source line 1361); the single entry cited inside the counted proof is reproduced verbatim in this task's own bibliography.tex, extracted from line 1417 of the pinned file and not edited in any way.

Editorial note. The counted proof has one reference to another labelled item, the definition labelled def:Stolz_groups, recorded as a navigation stub of kind definition.

Editorial note. Assembly history: r001 failed with 'Text line contains an invalid character' because the per-card bibliography file had been written through a shell heredoc that turned the leading backslash of \begin{thebibliography} into a backspace character, so the entry never registered and the citation stayed undefined; r002 rewrites that file with the file-writing tool instead of a shell script. The entry text itself was already correct and remains verbatim from line 1417 of the pinned source.

Allowed standard prerequisites (editorial).

Date: doi:10.3842/SIGMA.2025.093; CC BY 4.0.

- The three equivariant bordism groups appearing in the sequence, with their boundary, forgetful and closed-manifold maps. Same source, Definition labelled def:Stolz_groups; invoked in the counted statement and used throughout the counted proof. Fixes the objects and the three maps whose exactness is asserted.
- The classical non-equivariant Stolz exact sequence. Cited inside the counted proof (the long exact sequence (4.4) of Stolz); the author gives the equivariant argument in full. Locates the statement in the literature while the proof supplied is self-contained.

Source reference. Original definition 3.1: sources/geometry-fl04-stolz-positive-scalar-curvature-st/pinned-original-source.tex, source lines 368–368. Navigation pointer for \ref{def:Stolz_groups} used inside the counted span (line 368); the argument that consumes it is retained in the span.

Proposition 0.1. *The abelian groups defined in Definition* 3.1 *fit into the following G-equivariant version of the Stolz positive scalar curvature exact sequence:*

$$\cdots \to R^{\mathrm{spin}}_{n+1}(X)^G \to \mathrm{Pos}^{\mathrm{spin}}_n(X)^G \to \Omega^{\mathrm{spin}}_n(X)^G \to R^{\mathrm{spin}}_n(X)^G \to \cdots,$$

where the first map sends a manifold to its boundary, the second one is the forgetful map (i.e., it forgets the metric of positive scalar curvature) and the last one considers a closed manifold as a manifold with empty boundary.

Proof. This is a rather direct consequence of the definitions and well known to the experts, at least non-equivariantly, stated, e.g., in [1, long exact sequence (4.4)] (but without proof). The argument for the G-equivariant case is exactly the same as the classical non-equivariant situation. As we are not aware of a treatment available in the journal literature, we follow the referee's suggestion and give a complete account of the argument here.

First, that the composition of two consecutive maps is zero indeed is a direct consequence of the definitions:

- Given $[W, f, g] \in R^{\mathrm{spin}}_{n+1}(X)^G$, its image in $\Omega_n(X)^G$ is represented by $(\partial W, f|_{\partial W})$ which indeed is a boundary, hence represents 0.
- Given $[M, f, g] \in \mathrm{Pos}^{\mathrm{spin}}_n(X)^G$, its image in $R^{\mathrm{spin}}_n(X)^G$ is the manifold M with empty boundary. It is bordant in $R^{\mathrm{spin}}_n(X)^G$ to $\varnothing$ and hence represents 0 via the bordism $M \times [0,1]$, where $(M \times \{1\}, g)$ is a positive scalar curvature bordism of $\varnothing = \partial M$ to $\varnothing = \partial\varnothing$.
- Given $[M, f] \in \Omega^{\mathrm{spin}}_n(X)^G$, its image in $\mathrm{Pos}^{\mathrm{spin}}_{n-1}(X)^G$ is represented by $\partial M = \varnothing$, hence represents 0.

For the opposite inclusions of the kernels in the image, the constructions are almost as straight-forward:

- Assume that $[M, f] \in \Omega^{\mathrm{spin}}_n(X)^G$ is mapped to 0 in $R^{\mathrm{spin}}_n(X)^G$. This means that there is a null-bordism (W, F). In the case at hand, ∂W has two disjoint parts: on the one hand (M, f) and on the other hand $\big(M', f', g'\big)$ which is a bordism from $\partial M = \varnothing$ to $\partial\varnothing = \varnothing$ and which is equipped with a metric g of positive scalar curvature. But this means, by definition, that $[M, f] = \big[M', f'\big] \in \Omega^{\mathrm{spin}}_n(X)^G$ where M' is equipped with a metric g of positive scalar curvature. Hence we have the preimage $\big[M', f', g\big] \in \mathrm{Pos}^{\mathrm{spin}}_n(X)^G$ of the initial equivariant bordism class (M, f).
- Assume that $[M, f, g] \in \mathrm{Pos}^{\mathrm{spin}}_n(X)^G$ is mapped to 0 in $\Omega^{\mathrm{spin}}_n(X)^G$. This means that there is a bordism (W, F) with $\partial(W, F) = (M, f)$. But then $[W, F, g]$ represents a class in $R^{\mathrm{spin}}_{n+1}(X)^G$ which is mapped to $[M, f, g] \in \mathrm{Pos}^{\mathrm{spin}}_n(X)^G$.

- Finally, assume that $[W, f, g] \in R^{\mathrm{spin}}_{n+1}(X)^G$ is mapped to 0 in $\mathrm{Pos}^{\mathrm{spin}}_n(X)^G$. This means that there is a second bordism $\left(W', f', g'\right)$ whose boundary is $(\partial W, f|_{\partial W}, g)$. We now consider the closed G-manifold $Z := W \cup_{\partial W} W'$ with map $F := f \cup_{\partial W} f' \colon Z \to X$ and the bordism $B := Z \times [0,1]$ with map $F \circ \mathrm{pr}_Z \colon B \to X$. The boundary of B consists of three parts: the internal boundary W' between ∂W and $\varnothing = \partial Z$, equipped with the metric g' of positive scalar curvature extending g. The other two boundary parts are W and Z, and by definition $(B, F \circ \mathrm{pr}_Z, g')$ is a bordism in $R^{\mathrm{spin}}_n(X)^G$ between (W, f, g) and $(Z, f, \varnothing)$ and $[Z, f] \in \Omega^{\mathrm{spin}}_n(X)^G$ represents a preimage of $[W, f, g] \in R^{\mathrm{spin}}_n(X)^G$. □

References

[1] Stolz S., Concordance classes of positive scalar curvature metrics, Preprint, University of Notre Dame, 1999, https://web.archive.org/web/20240430033820/https://www3.nd.edu/~stolz/concordance.ps.